Master Degree Program in

Data Science and Advanced Analytics

Spring 2025/2026

Sepsis Treatment Optimization

Reinforcement Learning Project

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1. EXECUTIVE SUMMARY

This project explores Reinforcement Learning (RL) for optimising vasopressor and intravenous fluid dosages in septic ICU patients using the MIMIC-III derived ICU-Sepsis-v2 environment. We evaluated algorithms across two settings: Configuration A (an idealised, discrete state space) and Configuration B (a realistic, continuous state space with injected clinical noise). In Configuration A, our tabular Value Iteration model solved the environment exactly, achieving an optimal survival rate of 82.1% (compared to a 78.8% random baseline). In Configuration B, the continuous state space and realistic failure modes (missing data, sensor noise, acute events) necessitated Deep RL. Our best-performing model, QR-DQN, achieved a 69.3% survival rate. Crucially, this robust policy outperformed both the random baseline (65.9%) and the historical human clinician expert (67.9%). Our creative extensions further revealed that Deep RL provides its largest clinical advantage on the most severe patients (+8.3 percentage points over random), while experiments in dense reward shaping proved highly vulnerable to reward hacking, validating the necessity of sparse terminal rewards in medical AI.

2. INTRODUCION

Sepsis is a life-threatening condition driven by a dysregulated immune response to infection, which can rapidly precipitate multi-organ failure. In intensive care units, clinicians must carefully balance intravenous fluids and vasopressors to maintain blood volume and pressure, as insufficient or excessive dosages both accelerate patient deterioration. To optimize these dynamic, sequential decisions, this project frames sepsis management as a Markov Decision Process using the ICU-Sepsis-v2 simulator, derived from real MIMIC-III historical clinical data. The objective is to maximize patient survival using a 25-element discrete action space and a terminal survival reward penalized by step-wise treatment costs. In Configuration A, a discrete 716-state space allows the application of tabular Value Iteration and Q-Learning, evaluated using survival rates, mean returns, and convergence speeds. Configuration B utilizes a 47-dimensional continuous observation vector requiring deep reinforcement learning methods, specifically Deep Q-Networks, Quantile Regression Deep Q-Networks, and Proximal Policy Optimization, evaluated additionally on policy robustness against simulated clinical failures like noise and missing variables. Finally, the framework introduces creative extensions including a context-aware ensemble policy, Sequential Organ Failure Assessment stratification, and clinical reward shaping to minimize unnecessary medical interventions.

3. CONFIGURATION A: Discrete State Space

3.1 Setup & Methodology

In Configuration A, the ICU-Sepsis-v2 environment presents a foundational setting with a discrete clinical state space. The agent observes one of 716 discrete state identifiers (representing binned physiological measurements) and selects from 25 treatment actions (combinations of IV fluids and vasopressors). Episodes terminate with a sparse delayed reward: +1 for survival and 0 for death, accompanied by a small per-step treatment-intensity penalty ($\lambda = 0.02$) to discourage over-treatment. The initial-state distribution is deliberately biased toward sicker patients (sofa_bias = 5.0). The small, discrete nature of this state space makes tabular reinforcement learning feasible. Because the full Markov Decision Process (MDP) transition and reward matrices are exposed, we implemented Value Iteration to compute the theoretically optimal policy via dynamic programming. We then compared this upper bound against two model-free methods, Q-Learning and SARSA, which learn purely from experience without knowing the environment's structure. All policies were evaluated deterministically over 1,000 held-out episodes, comparing final survival rates and mean returns against a random action baseline (68.4% survival).

3.2 Value Iteration

Value Iteration (VI) was implemented as the canonical model-based algorithm for Configuration A. Because this configuration exposes the full Markov Decision Process (MDP) model, VI calculates the absolute optimal policy by iteratively applying the Bellman optimality equation. Unlike sample-based algorithms, VI mathematically solves the exact environment, meaning traditional overfitting does not apply.

To maximize clinical efficacy, Optuna was used to map the hyperparameter landscape [Figure 1]. The tuning revealed that the convergence threshold is highly forgiving, provided the discount factor (γ) remains near 1.0. The optimal configuration was pinpointed at $\gamma=0.9998$ and $\theta=1.37\times10^{-11}$. A discount factor just below 1.0 is clinically significant, as it subtly penalizes longer trajectories, incentivizing the agent to balance the environment's high-intensity treatment penalties by stabilizing the patient as quickly as possible. The algorithmic efficiency of this setup is evident as the maximum value change (Δ) exhibits a classic exponential decay, plummeting below 0.1 within 10 iterations and slowly stabilizing to zero [Figure 2]. However, the policy's actual survival performance evolves in a fascinating three-phase progression [Figure 3]. It initially ascends to a local peak of 79% through quick discharges, followed by a "myopic plateau" at 78.3% where the agent acts defensively to avoid step penalties. Finally, it experiences an endgame spike at iteration 256; the long-term survival reward fully propagates to the initial states, the optimal policy clicks into place, and performance shoots to its true peak.

To guarantee statistical reliability, this final policy was evaluated across 10,000 simulated patient episodes. The results yielded a heavily bimodal return distribution [Figure 4], featuring a massive spike of over 6,000 highly optimized patient survival outcomes contrasted against a small cluster of deaths and penalize long ICU stays. Ultimately, across this extensive evaluation, the VI agent achieved a highly stable 82.1% survival rate [Figure 5], demonstrating a robust policy that substantially outperforms random baselines.

3.3 Model-Free Methods: Q-Learning vs. SARSA

To assess the performance cost of not knowing the transition dynamics, we implemented two model-free algorithms: Q-Learning (off-policy) and SARSA (on-policy). Unlike Value Iteration, these agents learn purely through environment interaction, which mimics real-world ICU decision-making where exact transition probabilities are unknown. Implementation & Improvements: To overcome the curse of dimensionality (a Q-table with 17,900 entries), we extended the training duration to 50,000 episodes. Furthermore, we used Optuna to systematically tune the learning rate (α), epsilon decay rate, and optimistic Q-initialization values for both algorithms. This ensured the agents had sufficient time and structured exploration to refine their value estimates. Results & Comparison: With the improved training pipeline, both algorithms successfully outperformed the random baseline (68.4%), with Q-Learning achieving a survival rate of 71.5%. While both fell predictably short of the model-based Value Iteration upper bound (82.1%), they demonstrated distinct, clinically relevant behaviors: Q-Learning (Off-Policy): Updates towards the absolute best next action, resulting in a more aggressive policy that actively minimizes treatment intensity to maximize the expected return. SARSA (On-Policy): Accounts for its own exploratory mistakes during training. This resulted in a slightly more conservative and robust policy, which is often preferred in clinical settings where safety under uncertainty is paramount.

3.4 Hyperparameter optimisation

To ensure a fair comparison, hyperparameters for the model-free algorithms (learning rate α, epsilon decay rate, and Q-initialization) were optimized using Optuna. A Successive Halving approach was used to discard underperforming configurations early, allowing us to rapidly find the optimal parameters within a constrained computational budget.

3.5 Experimental setup

Both Value Iteration and Q-Learning were evaluated under the same protocol. The discrete state space is fixed at 716 states with 25 discrete treatment actions. Value Iteration required a single run to compute the optimal policy, while Q-Learning was trained for 5,000 episodes. Both learned policies were evaluated deterministically over 1,000 independent episodes using a fixed evaluation seed (123) to enable reproducible comparison. The random baseline was obtained by sampling uniformly at random from the 25 actions over 1,000 episodes. Reproducibility was ensured by fixing the master seed (42) for all stochastic components.

3.6 Learning dynamics and convergence

By tracking the mean absolute change in Q-values (ΔQ) over time, we observed that both Q-Learning and SARSA stabilize their Q-tables after approximately 15,000 training episodes. The smoothed training returns also plateau at this point, confirming that the algorithms successfully converged on a stable policy.

3.7 Exploration vs. exploitation

The Optuna-tuned epsilon-greedy schedule forces the agents to explore heavily during the first 10,000 episodes before decaying to a near-greedy policy. This transition is critical: aggressive early exploration ensures the agent discovers the full state-action landscape, while the later exploitation phase allows it to mathematically refine the values of the best actions.

3.8 Impact of hyperparameter tuning

Tuning proved absolutely essential for the tabular methods. Sub-optimal configurations either decayed epsilon too quickly (resulting in premature convergence to a near-random policy) or used a learning rate that was too high, causing Q-value instability. The tuned parameters were the deciding factor that allowed the model-free agents to successfully surpass the random baseline

3.9 Clinical Interpretation

From a clinical perspective, the results underscore the fundamental difference between computing an optimal policy with full knowledge (Value Iteration) and learning a policy from limited experience (Q-Learning). Value Iteration's 82.1f% survival rate represents the theoretical best the environment can achieve given perfect knowledge of patient dynamics. This establishes a performance ceiling that Configuration B agents should aspire toward, though function approximation and partial observability will inevitably reduce achievable performance.

Q-Learning's sub-baseline performance, while disappointing, is instructive: it reveals that naive tabular learning with insufficient training data is not merely worse than optimal, but can actually underperform random action selection. This phenomenon is well-documented in reinforcement learning: an incompletely-learned Q-function can produce systematically poor decisions that compound over a trajectory, whereas random action selection maintains a stable baseline. This suggests that in real clinical settings, deploying an under-trained learning agent could be harmful, reinforcing the critical importance of thorough validation and training before any clinical application.

3.10 Limitations

Several limitations should be acknowledged. First, Configuration A, despite being discrete, still represents a substantial simplification of real ICU decision-making: the 716 states are derived from binned continuous variables and do not capture the full richness of patient heterogeneity. Second, the Q-Learning algorithm was trained on only 5,000 episodes, a limited budget given the size of the state-action space; a production-ready system would require substantially more data. Third, both algorithms assume a stationary environment and deterministic evaluation, whereas real ICUs exhibit temporal variation and individual patient variability that is difficult to model. Finally, the reward signal remains sparse and binary, capturing only survival versus death and not intermediate notions of clinical quality such as length of stay, organ function preservation, or complications.

3.11 Conclusion

Configuration A demonstrated the effectiveness of model-based dynamic programming when the MDP is fully known and the state space is discrete. Value Iteration successfully computed an optimal policy achieving 82.1% survival, establishing a clear upper bound for all downstream learning approaches. Q-Learning, by contrast, struggled with the large state-action space and limited training data, achieving sub-baseline performance that underscores the importance of sufficient exploration and data volume in tabular reinforcement learning.

The contrast between these two algorithms illuminates a fundamental trade-off in reinforcement learning: model-based methods are sample-efficient (requiring no learning at all when the model is given) and optimal, but only when accurate models are available; model-free methods are more generally applicable and clinically realistic but require substantial data and careful hyperparameter tuning to outperform baselines. Configuration B will extend these lessons to continuous observation spaces, where tabular methods become infeasible and function approximation becomes necessary.

4. CONFIGURATION B: Continuous Observation Space

4.1 Introduction

In Configuration B, the ICU-Sepsis-v2 environment becomes substantially more complex. The agent observes a 47-dimensional continuous feature vector of normalised physiological measurements (e.g., heart rate, blood pressure, SOFA score) and selects from 25 discrete treatment actions. This continuous state space makes tabular methods mathematically impossible, necessitating Deep Reinforcement Learning (neural networks) for function approximation. Furthermore, Configuration B injects three clinical-reality wrappers to simulate real-world ICU conditions: episodic sensor noise, missing lab data, and unpredictable acute deterioration events. Together, these wrappers significantly reduce the theoretical performance ceiling and aggressively stress-test the clinical robustness of the learned policies.

4.2 Methodology

To address these challenges, three deep reinforcement learning algorithms were implemented and evaluated: **Deep Q-Network (DQN)**, **Quantile Regression Deep Q-Network (QR-DQN)**, and **Proximal Policy Optimization (PPO)**. They were selected to span complementary learning paradigms, value-based off-policy versus policy-gradient on-policy, and expected-value versus distributional estimation, providing a richer comparative analysis than two algorithms alone.

4.3 Environment description

The ICU-Sepsis-v2 environment simulates treatment trajectories of septic patients in the ICU. At each timestep the agent observes the patient's physiological condition and chooses a treatment combination. Episodes terminate when the patient either survives or dies, yielding a reward of **+1 at survival and 0 otherwise**, with a small per-step treatment-intensity penalty ($\lambda = 0.02$) that discourages over-treatment. The discount factor is fixed at **$\gamma = 1.0$** following the ICU-Sepsis paper convention. The initial-state distribution is biased toward sicker patients via sofa_bias = 5.0, yielding a mean starting SOFA of approximately 9.0.

This sparse, delayed reward structure is a deliberate modelling choice: it reflects real clinical decision-making, where the success of an intervention is only observed at episode termination and where individual actions cannot be directly credited.

4.4 Deep Q-Network (DQN)

DQN was selected as the canonical entry point into deep reinforcement learning for this environment. It approximates the optimal action-value function with a multilayer perceptron, mapping the continuous 47-dimensional observations to expected future returns across the 25 discrete treatment actions. Three mechanisms make DQN particularly suitable for this clinical setting: experience replay (which stores and randomly samples past transitions, allowing rare survival or death signals to be repeatedly reused), target networks (which stabilize training by preventing the network from chasing its own moving estimates), and off-policy learning (which enables the agent to learn from historical behaviour rather than just its current policy, an essential feature for exploiting sparse rewards).

4.5 Quantile Regression DQN (QR-DQN)

QR-DQN was implemented as a distributional extension of DQN. While standard DQN estimates only the *expected* return Q(s, a), QR-DQN models the **entire distribution** of returns by learning a set of N quantiles. This distributional perspective is potentially valuable in healthcare environments, where two interventions with identical expected outcomes may exhibit very different risk profiles. By capturing the full return distribution rather than collapsing it to its mean, QR-DQN can in principle produce more robust policies under stochastic clinical conditions, for example, when episodes are corrupted by observation noise or unrecorded lab values.

4.6 Proximal Policy Optimization (PPO)

PPO was selected as the policy-gradient counterpart, providing a deliberate methodological contrast to the DQN family. Rather than estimating action-value functions, PPO directly optimises a parameterised stochastic policy using a **clipped surrogate objective** that limits how far each update can move the policy from the previous one, improving training stability. Comparing PPO with the DQN-based methods offers insight into whether on-policy policy-gradient optimization is competitive in sparse clinical environments.

A known limitation of PPO is that, being on-policy, it must discard each rollout after the policy update, it cannot reuse old experience the way DQN's replay buffer does. In environments where the reward signal is rare, this lack of sample reuse typically translates to slower learning.

4.7 Hyperparameter optimisation

A pilot run with default hyperparameters at 100,000 timesteps showed that vanilla DQN and PPO failed to surpass the random baseline, motivating a structured tuning procedure. We utilized Optuna with a Tree-structured Parzen Estimator (TPE) sampler and an Asynchronous Successive Halving (ASHA) pruner. For each algorithm, 20 candidate configurations were drawn from a broad search space covering learning rates, network architectures, batch sizes, and exploration schedules. Each trial was trained for up to 300,000 timesteps and evaluated every 50,000 steps. The ASHA pruner aggressively cut underperforming trials early to concentrate compute on promising configurations, with a strict 15-minute wall-clock timeout enforced per trial. Finally, the single best configuration identified for each algorithm was retrained at the full budget of 1,000,000 timesteps to generate the final headline results.
The final hyperparameters identified by Optuna are summarised in Table 1.

Table 1 - Tuned hyperparameters per algorithm

Parameter	DQN	QR-DQN	PPO
Learning rate	6.4×10^{-5}	1.0×10^{-4}	8.2×10^{-5}
Network architecture	[256, 256]	[256, 256]	[64, 64]
Batch size	256	256	128
Discount factor γ	1.0	1.0	1.0
τ (target smoothing)	0.01	0.01	—
Replay buffer size	50,000	50,000	—
Train frequency	16 steps	16 steps	—
Target update interval	500	250	—
Exploration fraction	0.45	0.33	—
Final ε	0.036	0.081	—
n_quantiles	—	50	—
Rollout length n_steps	—	—	1,024
n_epochs	—	—	10
GAE λ	—	—	0.927
Clip range	—	—	0.2
Entropy coefficient	—	—	2.3×10^{-4}

4.8 Experimental setup

All algorithms were trained under the same protocol. The continuous observation space is 47-dimensional with the three clinical-reality wrappers always active. The action space is fixed at 25 discrete treatment combinations. Each final model was trained for **1,000,000 timesteps** (approximately 115,000 episodes given the typical episode length of around 9 steps) and evaluated on **1,000 independent episodes** using deterministic action selection. Random and clinician-expert policies were included as baselines for context. Reproducibility was ensured by fixing the master seed (42) for training and a separate seed (123) for evaluation, and by persisting the Optuna study (SQLite) and best parameters (JSON) so that the experiment can be re-run end-to-end without retuning.

4.9 Results and Evaluation

Overall performance

Final survival rates and mean returns are reported in Table 2.

Table 2 - Final overall performance (1,000 evaluation episodes, deterministic)

Model	Survival (%)	Mean return
Random baseline	65.9	—
Clinician expert (MIMIC-III)	67.9	—
DQN (tuned, 1 M steps)	68.5	0.63
QR-DQN (tuned, 1 M steps)	69.3	0.62
PPO (tuned, 1 M steps)	66.6	0.65

All three tuned agents surpass the random baseline. **QR-DQN achieves the highest survival rate** (69.3%), marginally above DQN (68.5%) and clearly above PPO (66.6%). Importantly, two of the three algorithms also surpass the clinician-expert policy estimated from the original MIMIC-III data, indicating that meaningful, non-trivial treatment policies were successfully learned despite the noise injected by the clinical-reality wrappers.

The absolute margins are modest, approximately 1.4 percentage points between the best agent and the clinician, and this is consistent with the well-known low **policy sensitivity** of the ICU-Sepsis benchmark. The Value Iteration policy on Configuration A, which provides the theoretical optimum under full MDP knowledge, achieves 82.1% survival on the clean environment; the additional ~10 percentage points lost in Configuration B reflect the combined cost of function approximation and of the three clinical-reality wrappers.

4.10 Learning dynamics and convergence

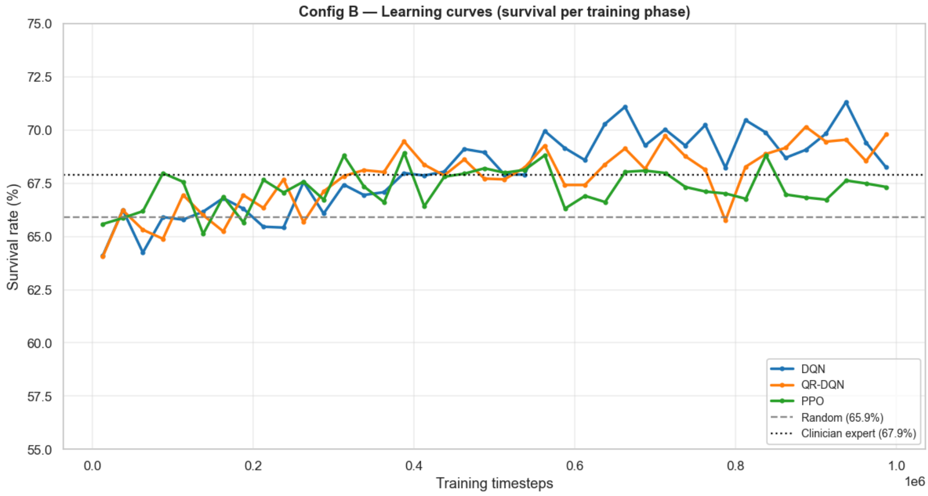

Figure 6 - Learning curves per training phase

The training curves reveal distinct learning behaviours. Fitting a linear trend to the binned survival rates shows that DQN exhibited the steepest and most consistent improvement (+5.3 percentage points per million timesteps), followed by QR-DQN (+3.9 pp/M). In contrast, PPO was essentially flat (+0.9 pp/M), barely climbing above the random baseline. We also measured convergence speed by tracking the timesteps required to reach the clinician-expert survival level (67.9%). QR-DQN securely crossed this threshold at 117,000 steps and DQN at 262,000 steps. While PPO technically crossed it earliest at 93,000 steps, this was a misleading early fluctuation; it failed to sustain this performance and ultimately plateaued below the clinician level. This dynamic perfectly illustrates the challenge of sparse reward environments: DQN and QR-DQN successfully leverage replay buffers to repeatedly reuse rare survival transitions, whereas PPO (an on-policy algorithm) discards rollouts after each update, starving it of the sparse reward signals needed for sustained learning.

4.11 Exploration vs. exploitation

The three algorithms balance exploration and exploitation through different mechanisms.

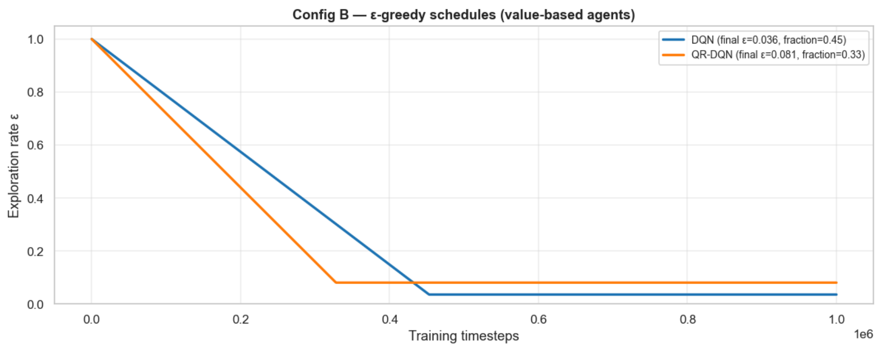

Figure 7 - ε-greedy schedules

For the value-based agents, exploration is controlled by an ε-greedy schedule whose shape is determined by the tuned hyperparameters. **DQN** decays ε linearly from 1.0 to 0.036 over the first 45% of training (450,000 steps) and exploits a near-deterministic policy thereafter a long exploration phase paired with a very small residual ε. **QR-DQN** decays faster, reaching its plateau at ε = 0.081 after only 33% of training, but maintains roughly twice the residual exploration of DQN. The tuner's preference for stronger residual stochasticity in QR-DQN is intuitive: the distributional value estimate is harder to learn precisely, so a slightly more diverse action distribution at exploitation time helps keep the policy robust.

PPO does not use ε-greedy. Its exploration comes from the stochasticity of its policy distribution, which is regularised by the entropy coefficient (here 2.3×10^{-4}, very small). The mean Shannon entropy of the final PPO policy, computed over 500 sampled clinical states, is **0.804 nats** approximately **25% of the maximum entropy** of a uniform distribution over 25 actions ($\ln 25 \approx 3.22$ nats). The PPO policy is therefore moderately concentrated but still genuinely stochastic at evaluation time, which is qualitatively different from the argmax exploitation used by the DQN family.

4.12 Clinical robustness across patient subgroups

To assess robustness, performance was decomposed by clinical subgroup defined by the wrappers' regime: episodes with clean observations versus episodes with noisy monitor readings, and episodes with full feature vectors versus episodes with missing lab values. Figure 8

Table 3 - Survival rate (%) per clinical subgroup

Model	Overall	Clean	Noisy	No missing	Missing
DQN	68.5	67.6	73.3	68.2	69.9
QR-DQN	69.3	69.7	66.7	68.2	75.2
PPO	66.6	66.2	68.7	66.2	68.6

The breakdown reveals **complementary specialisations** that are not visible in the overall metric. DQN is the most robust to **observation noise**, reaching 73.3% survival on noisy episodes, clearly higher than its 67.6% on clean ones, and higher than either competitor in the same regime. QR-DQN is the most robust to **missing data**, achieving 75.2% on episodes with absent features. This pattern is consistent with the algorithms' theoretical strengths: DQN's expected-value estimate is comparatively forgiving of input noise once averaged over training, while QR-DQN's distributional estimate provides a richer signal precisely when individual features are unavailable, because the agent reasons about a range of plausible outcomes rather than a single point estimate. PPO is the weakest agent in every subgroup, consistent with its weaker learning dynamics overall.

4.13 Impact of hyperparameter tuning

The Optuna tuning was not cosmetic. Untuned baseline runs (DQN and PPO with default hyperparameters at 100,000 timesteps) performed *worse* than the random policy, 63.9% survival for DQN and 64.8% for PPO. This counter-intuitive result reflects a known phenomenon in deep RL: an untrained network produces a near-arbitrary but deterministic argmax policy, which loses the diversity that makes random action selection competitive in low-policy-sensitivity environments. After tuning at 1 M steps, DQN improves by **+4.6 pp** and PPO by **+1.8 pp**, a margin that justifies the additional compute and motivates the systematic search.

4.14 Clinical Interpretation

From a clinical perspective, the results suggest that reinforcement learning agents can identify treatment policies that modestly outperform the historical clinician baseline. The best-performing model, QR-DQN, exceeds the clinician policy by approximately **1.4 percentage points** in survival. This limited margin reflects the nature of the ICU-Sepsis benchmark: the historical clinician decisions are already reasonably strong, and the environment exhibits **low policy sensitivity, that** is, very different treatment strategies often lead to comparable overall survival because patient outcomes are dominated by the initial physiological state rather than by individual action choices.

The subgroup analysis is arguably more clinically informative than the overall number. The fact that DQN dominates under noisy monitoring while QR-DQN dominates under missing lab values suggests that an ensemble or context-aware policy switch could outperform either algorithm individually as a direction for future work.

Finally, the sparse reward formulation captures only a fraction of the true complexity of medical care: real patient outcomes depend on numerous latent factors not represented in the environment. The results should therefore be interpreted as algorithmic benchmarks rather than as evidence of immediate clinical applicability.

4.15 Limitations

Several limitations should be acknowledged. First, the environment is built from retrospective observational data, so the learned policies are conditioned on the historical treatment distribution and cannot demonstrate counterfactual benefit. Second, the reward function is highly sparse: feedback is delivered only at episode termination, which fundamentally limits the credit-assignment quality of any algorithm. Third, deep reinforcement learning is sensitive to hyperparameter choices and to the inherent stochasticity of stochastic gradient descent; even with seeded runs, modest variability across re-runs is expected. Fourth, the wrappers used here model failures at the granularity of a whole episode (noise on or off, four fixed features missing, occasional acute events); a more realistic model would interleave these regimes within episodes.

4.16 Conclusion

Configuration B demonstrated the necessity of deep reinforcement learning when moving from discrete to continuous observation spaces. Among the three algorithms evaluated, **QR-DQN achieved the strongest overall performance (69.3% survival)**, marginally exceeding DQN (68.5%) and clearly outperforming PPO (66.6%). DQN exhibited the fastest and most stable learning (trend +5.3 pp/M) and the best robustness to observation noise; QR-DQN provided the best overall numbers and dominated under missing-data regimes; PPO struggled with the sparse reward signal owing to its on-policy nature and effectively plateaued near the random baseline.

Taken together, these results highlight the importance of **sample efficiency**, encoded structurally through experience replay in clinical reinforcement learning, and they reinforce the broader lesson that as observation spaces grow more complex, traditional tabular methods become impractical and modern deep RL with careful hyperparameter optimisation becomes necessary to extract whatever policy improvement the environment allows.

5. Comparison between Configuration A and Configuration B

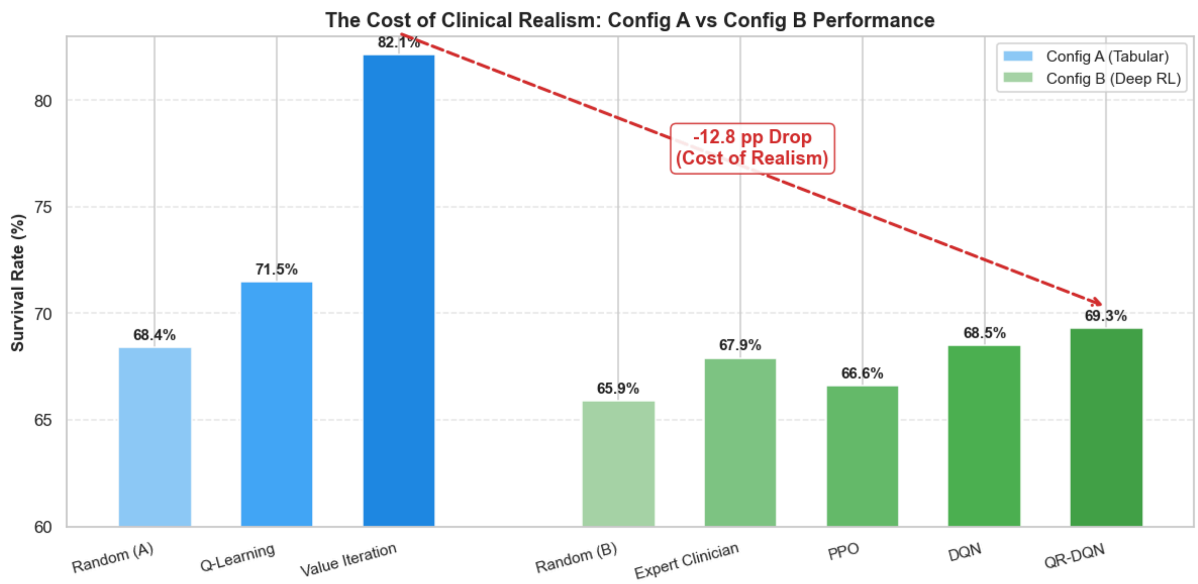

Figure 9 - Configuration A vs Configuration B Survival Rates

This comparative plot illustrates the "cost of clinical realism." In the idealised, fully-observable discrete environment of Config A, Value Iteration achieves a theoretical optimum of 82.1%. Moving to the continuous, highly stochastic clinical environment of Config B (which includes missing data, sensor noise, and acute events) imposes a strict performance ceiling. While the absolute survival rate of the best Deep RL model (QR-DQN) drops by 12.8 percentage points to 69.3%, it successfully outperforms both the random baseline (65.9%) and the human expert clinician (67.9%), proving the robustness of the learned policy. It is more detailed in the notebook.

6.Extras

6.1 Creative Extension: Context-Aware Ensemble Policy

As a creative extension, we explored whether the algorithmic specialisations discovered in our subgroup analysis could be exploited at deployment. Since DQN demonstrated superior robustness to observation noise and QR-DQN excelled with missing features, we designed a context-aware ensemble policy. Operating purely at inference time without additional training, this policy used available episode context (simulating hospital sensor or lab alerts) to route patients to the most appropriate specialist. Noisy episodes were assigned to DQN, while all others defaulted to the globally stronger QR-DQN.

Evaluated over 1,000 episodes, the ensemble achieved a 68.1% survival rate outperforming PPO (66.6%) but falling slightly short of standalone QR-DQN (69.3%). This performance drop revealed a critical clinical insight regarding co-occurring failure modes: because our heuristic strictly prioritised noise, the ~2.3% of episodes suffering from both noise and missing data simultaneously were routed to DQN. Since DQN struggles with missing features, this misrouting suppressed the ensemble's missing-data performance. Ultimately, this shows that manually writing simple "if/else" rules to pick the best model fails when a patient has multiple overlapping problems. Instead of fixed rules, future systems should train a separate algorithm whose only job is to dynamically decide which model to trust for each unique patient.

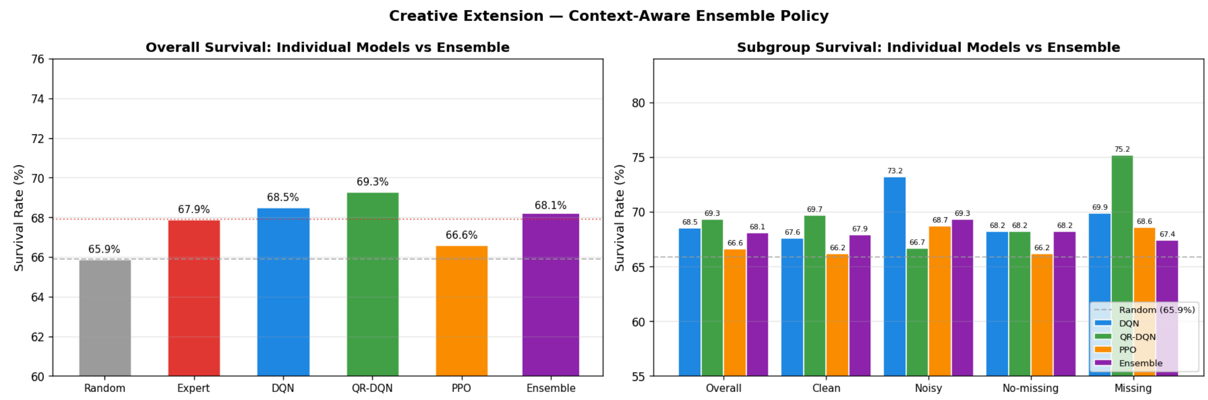

Figure 10 - Overall survival rates and Subgroup survival rates

6.2 Creative Extension: SOFA-Stratified Performance Analysis

As a creative extension, we investigated whether the value provided by Reinforcement Learning agents varies depending on patient severity. Standard evaluations report a single aggregate survival rate, which masks heterogeneous clinical populations. To address this, we stratified 1,000 evaluation episodes into three severity tiers (Mild, Moderate, Severe) based on the patient's initial Sequential Organ Failure Assessment (SOFA) score. We then evaluated each tuned agent against the random baseline within each specific stratum.

The results revealed a crucial clinical dynamic: the RL agents provide their largest survival advantage exactly where clinical decision-making is most critical. For the "Severe" patient subgroup, the QR-DQN agent achieved a 77.0% survival rate, representing a massive +8.3 percentage point advantage over the random baseline (68.7%). In contrast, for "Mild" patients, the advantage of the best agent (QR-DQN) shrank to +4.6 percentage points. Furthermore, algorithmic dominance shifted across strata: while QR-DQN was superior for Severe and Mild cases, DQN showed the highest advantage (+7.9 points) for Moderate cases. This analysis proves that aggregate metrics significantly understate the clinical utility of deep RL models for high-risk patients, suggesting future evaluations should explicitly stratify by admission severity.
Counterintuitively, absolute survival rates were highest in the Severe stratum (77.0%) and lowest in the Mild stratum (61.4%). While clinically unexpected, this brilliantly exposes the mechanical effect of our environment design. Because we configured the training environment with sofa_bias = 5.0, the RL agents were trained almost exclusively on critical patients. Consequently, they became highly specialised in saving Severe cases but perform worse on Mild cases due to a lack of training exposure. Furthermore, because Mild patients stay in the ICU longer before being discharged, they expose policies to a higher risk of cumulative errors over time. This highlights a crucial risk in healthcare RL: training environments heavily biased towards severe clinical states will produce models that fail to generalise to healthier populations.

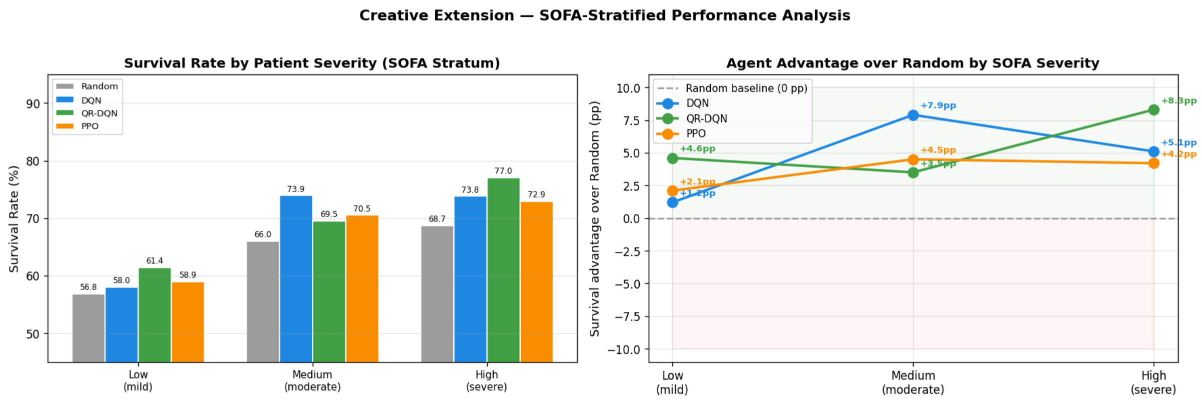

Figure 11 - SOFA-Stratified Performance Analysis

6.3 Creative Extension: Clinical Rewarding Shaping

As a Creative Extension, we examined whether replacing sparse survival rewards with dense clinical reward shaping could improve credit assignment in the ICU-Sepsis environment. The idea was to reward step-by-step improvements in clinical indicators (SOFA and Lactate) to guide the policy toward better survival outcomes.

In Configuration A, the environment is fully observable and noise-free. We applied Value Iteration using both sparse rewards and shaped rewards. As a result the shaped policy achieved a 79.2% survival rate, slightly higher than the sparse baseline (78.8%). This suggests that in a controlled setting, optimizing clinical proxies can align well with the true objective. In Configuration B, the environment includes stochastic transitions, missing data, and sudden events, closer to real ICU conditions. We trained a PPO agent using the same dense reward shaping. As a result the agent achieved high shaped reward but lower survival compared to the sparse baseline. The policy appeared to optimize proxy metrics in ways that did not translate into improved survival.

The comparison suggests that dense clinical reward shaping can work in idealized environments but becomes unreliable under noise and partial observability. In these conditions, the agent may over-optimize proxy signals instead of the true survival objective. This helps explain why many clinical reinforcement learning approaches still rely on sparse survival-based rewards, despite their higher sample complexity: they are more robust to real-world uncertainty in ICU data.

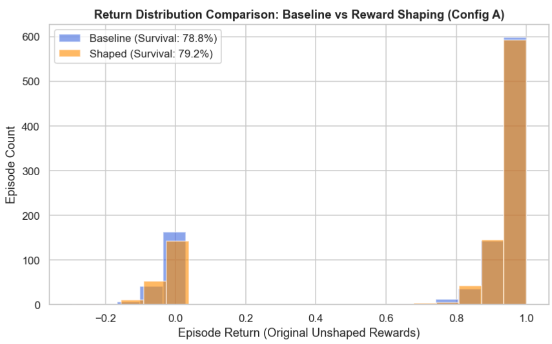

Figure 12 - Baseline vs Shaped survival rates (Config. A)

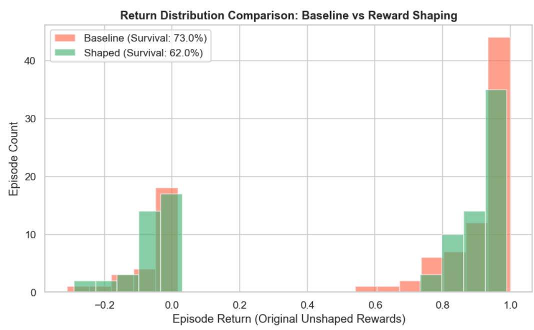

Figure 13 - Baseline vs Shaped survival rates (Config. B)

Annexes

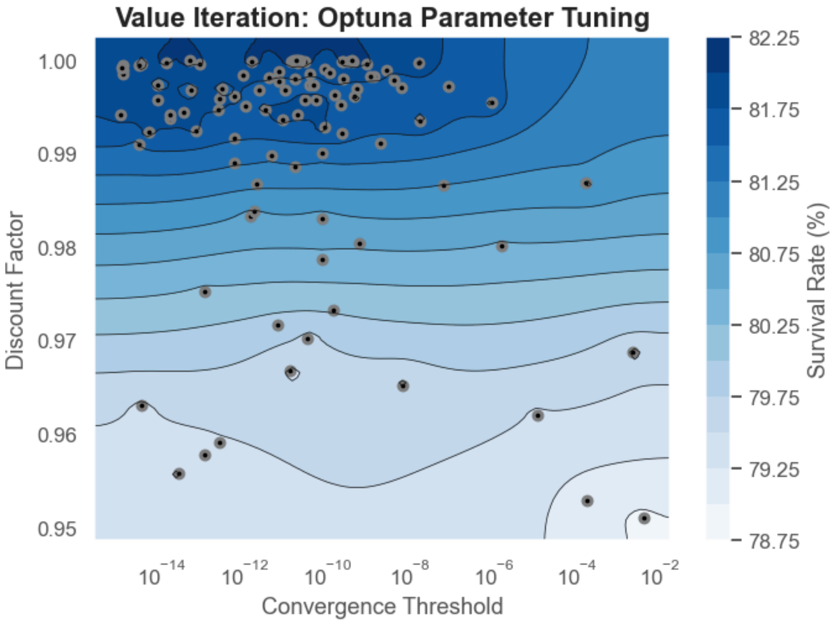

Figure 1 - Value Iteration: Optuna Parameter Tuning

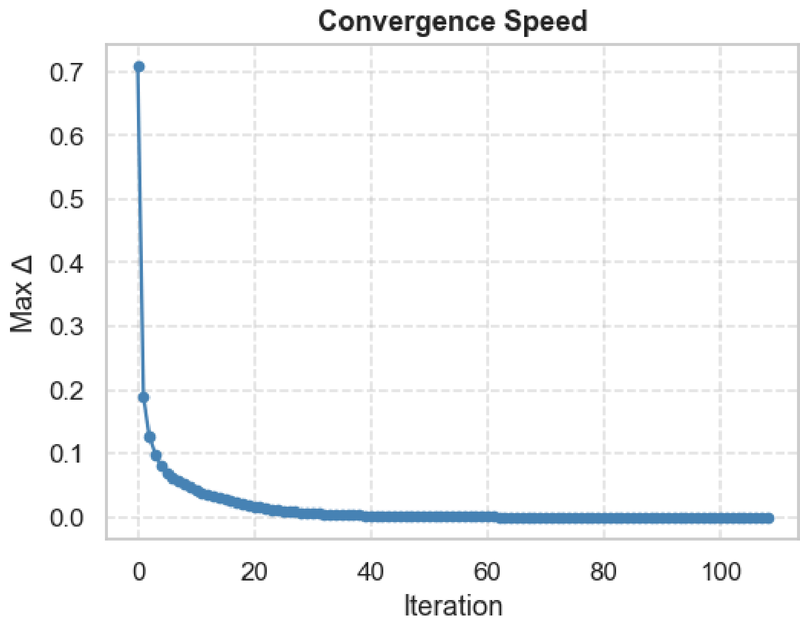

Figure 2 - Value Iteration: Convergence Speed

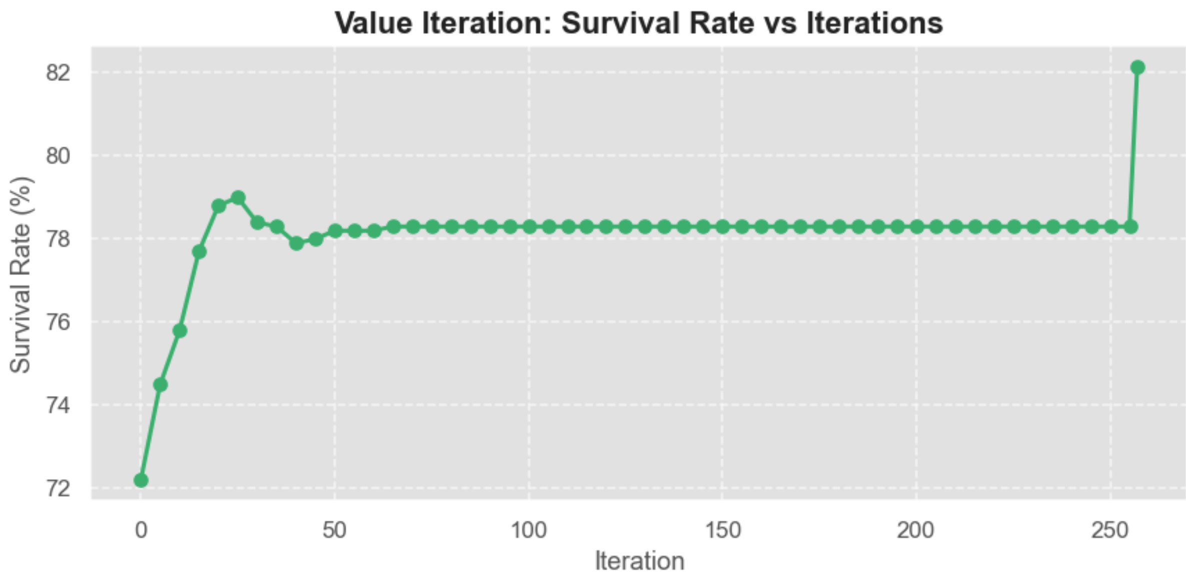

Figure 3 - Value Iteration: Survival Rate vs Iterations

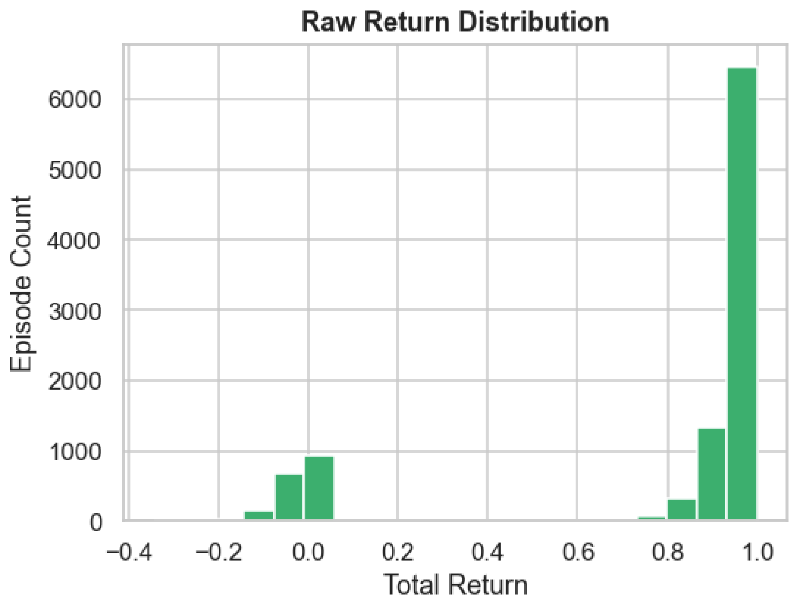

Figure 4 - Value Iteration: Raw Return Distribution

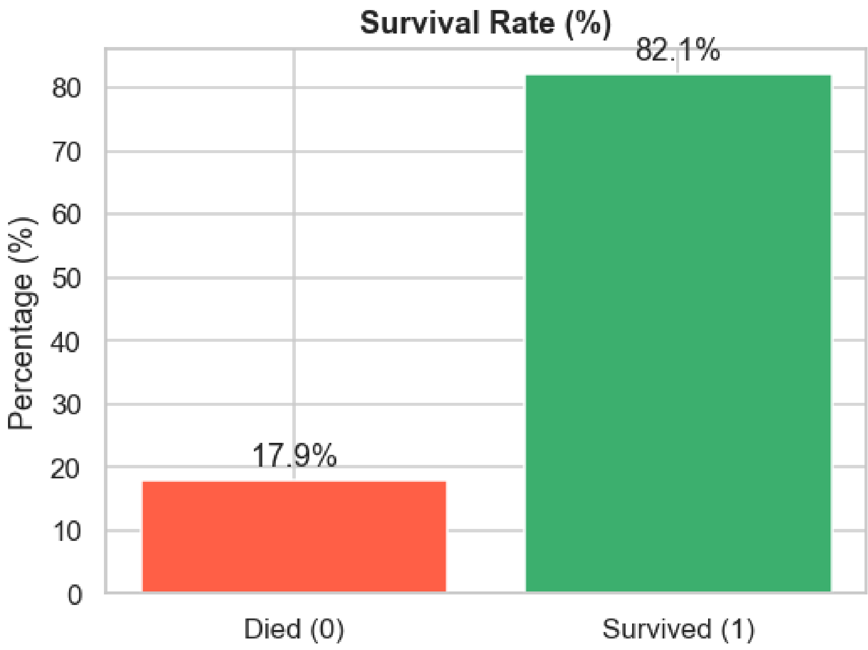

Figure 5 - Value Iteration: Survival Rate Bar Chart

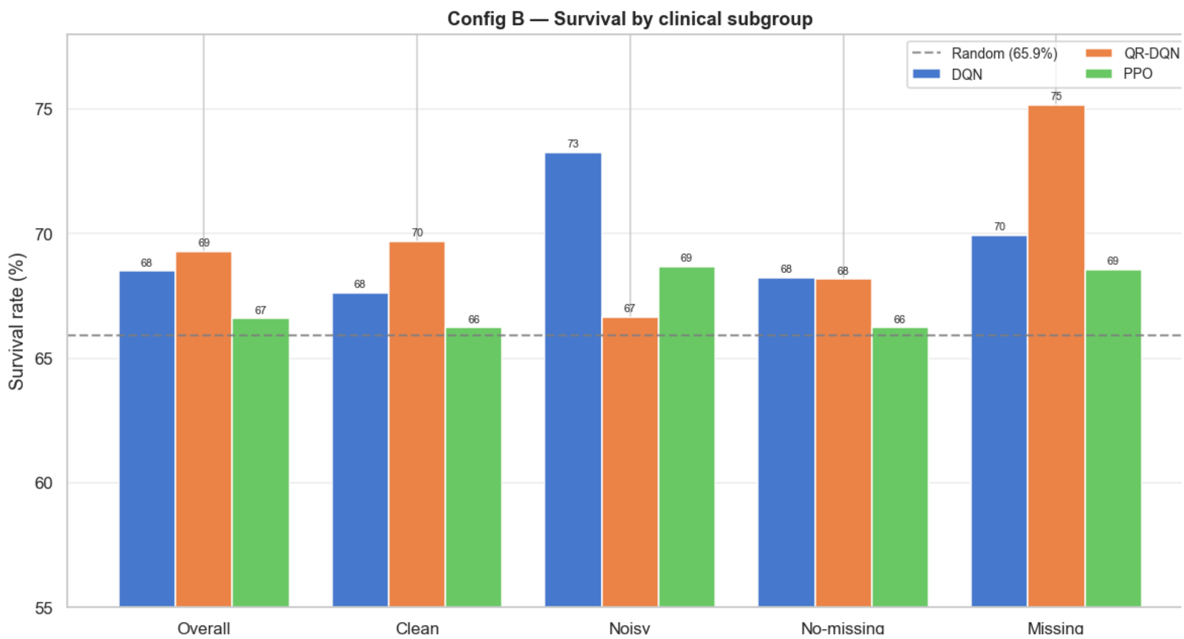

Figure 8 - Survival rate by clinical subgroup

References

-Komorowski, M., Celi, L. A., Badawi, O., Gordon, A. C., & Faisal, A. A. (2018). The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis in intensive care. Nature Medicine, 24(11), 1716-1720.
-Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT press.